

Grade 8
Unit 4
Lesson 3 Practice

Name Answer key
Date _____
Period _____

Apply the laws of exponents to write equivalent numerical expressions with one base and one positive exponent.

$$1. \quad \frac{5^{-5}}{5^{-7}} \cdot 5^4 = \frac{5^7}{5^5} \cdot 5^4 = 5^2 \cdot 5^4 = 5^6$$

$$2. \quad \frac{((-2)^4)^3}{(-2)^2 \cdot (-2)^{-2}} = \frac{(-2)^{12}}{(-2)^0} = (-2)^{12}$$

$$3. \quad \frac{(1.3^4)^3}{(1.3^2)^2} = \frac{(1.3)^{12}}{(1.3)^4} = (1.3)^8$$

$$4. \quad \left(\left(\frac{1}{3} \right)^0 \cdot \left(\frac{1}{3} \right)^{-3} \right)^2 = 1 \cdot \left(\frac{1}{3} \right)^{(-3)(2)} = \left(\frac{1}{3} \right)^{-6} = 3^6$$

$$5. \quad \left(\frac{(4)^2 \cdot (4)^3}{(4)^{-2}} \right)^{-2} = \frac{(4^5)^{-2}}{[(4)^{-2}]^{-2}} = \frac{(4)^{-10}}{(4)^4} = \frac{1}{4^4 \cdot 4^{10}} = \frac{1}{4^{14}}$$

Consider the expression $\frac{\left(\frac{1}{2}\right)^3}{2^{-6}} \cdot 2^6$. For each expression in questions 6-10, determine if it is equivalent to $\frac{\left(\frac{1}{2}\right)^3}{2^{-6}} \cdot 2^6$. Then, justify your choice by explaining using laws of exponents or showing your work.

$$\frac{\left(\frac{1}{2}\right)^3}{2^{-6}} \cdot 2^6 = \left(\frac{1}{2}\right)^3 \cdot 2^6 \cdot 2^6 = 2^3 \cdot 2^6 = 2^9$$

Expression	Is It Equivalent to $\frac{\left(\frac{1}{2}\right)^3}{2^{-6}} \cdot 2^6$?	Justify or Explain
6. $\frac{\left(\frac{1}{2}\right)^3}{\left(\frac{1}{2}\right)^6} \cdot 2^6$	Yes	$\left(\frac{1}{2}\right)^{-3} \cdot 2^6 = 2^3 \cdot 2^6 = 2^9$
7. 2^{15}	No	$2^{15} \neq 2^9$
8. $\frac{(2)^3}{(2)^6} \cdot 2^6$	No	$2^{-3} \cdot 2^6 = 2^3$
9. 2^9	Yes	$2^9 = 2^9$
10. $\frac{(2)^{-3}}{(2)^{-6}} \cdot 2^6$	Yes	$2^{-3+6} \cdot 2^6 = 2^3 \cdot 2^6 = 2^9$